

Assignment 4

INFS-343

Part I: Complete this part using your own R work. Do not use AI tools for this section.

1. First impression

What is your first impression on this csv datafile? Did you notice anything odd? Any missing values? Any extreme values? Please write down your first impression here.

This is the first step of a data analysis.

My first impression is that the dataset is mostly clean and easy to use. Most variables do not have missing values. However, I noticed that the **Cabin** variable has many missing values, so it may not be very useful. The **Fare** variable has only one missing value. I also saw some extreme values. For example, the **Fare** goes from 0 to 512, which is a very big difference. This means there are some outliers.

The most unusual thing is that **all females survived and all males did not survive**. This is very strange and not realistic compared to real Titanic data. Because of this, I expect **sex** to be the most important variable in the models.

2. Descriptive statistics

Now load the data into R, provide some descriptive analysis on the variables.

```
> summary(titanic)
  PassengerId   Survived  Pclass     Sex       Age
Min.   : 892.0   Min.   :0.0000   Min.   :1.000   Length:332   Min.   : 0.17
1st Qu.: 992.8   1st Qu.:0.0000   1st Qu.:1.000   Class :character   1st Qu.:21.00
Median :1099.5   Median :0.0000   Median :2.000   Mode  :character   Median :27.00
Mean   :1100.1   Mean   :0.3825   Mean   :2.145                Mean   :30.27
3rd Qu.:1210.2   3rd Qu.:1.0000   3rd Qu.:3.000                3rd Qu.:39.00
Max.   :1307.0   Max.   :1.0000   Max.   :3.000                Max.   :76.00

  SibSp   Parch   Ticket   Fare       Cabin
Min.   :0.0000   Min.   :0.0000   Length:332   Min.   : 0.00   Length:332
1st Qu.:0.0000   1st Qu.:0.0000   Class :character   1st Qu.: 8.05   Class :character
Median :0.0000   Median :0.0000   Mode  :character   Median :16.00   Mode  :character
Mean   :0.4819   Mean   :0.3976                Mean   :40.98
3rd Qu.:1.0000   3rd Qu.:1.0000                3rd Qu.:40.63
Max.   :8.0000   Max.   :6.0000                Max.   :512.33
NA's   :1

  Embarked
Length:332
Class :character
Mode  :character

> str(titanic)
'data.frame': 332 obs. of 11 variables:
 $ PassengerId: int 892 893 894 895 896 897 898 899 900 901 ...
 $ Survived   : int 0 1 0 0 1 0 1 0 1 0 ...
 $ Pclass     : int 3 3 2 3 3 3 2 3 3 ...
 $ Sex        : chr "male" "female" "male" "male" ...
 $ Age        : num 34.5 47 62 27 22 14 30 26 18 21 ...
 $ SibSp      : int 0 1 0 0 1 0 0 1 0 2 ...
 $ Parch      : int 0 0 0 0 1 0 0 1 0 0 ...
 $ Ticket     : chr "330911" "363272" "240276" "315154" ...
 $ Fare       : num 7.83 7 9.69 8.66 12.29 ...
 $ Cabin      : chr "" "" "" "" ...
 $ Embarked   : chr "Q" "S" "Q" "S" ...

> colSums(is.na(titanic))
PassengerId   Survived   Pclass     Sex       Age       SibSp   Parch   Ticket
0              0          0          0          0          0          0
Fare          Cabin      Embarked
1              0          0
```

Please make sure you describe each variable briefly in the word document:

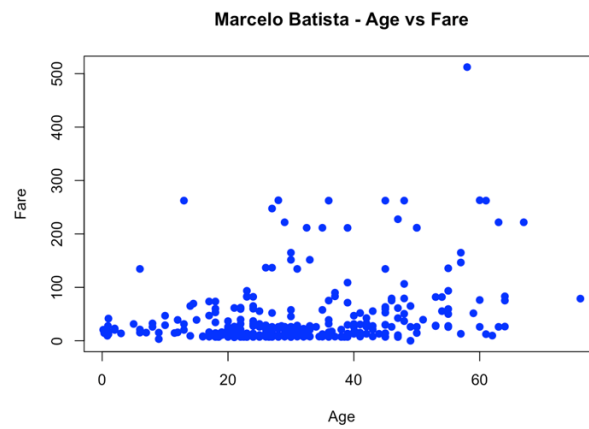
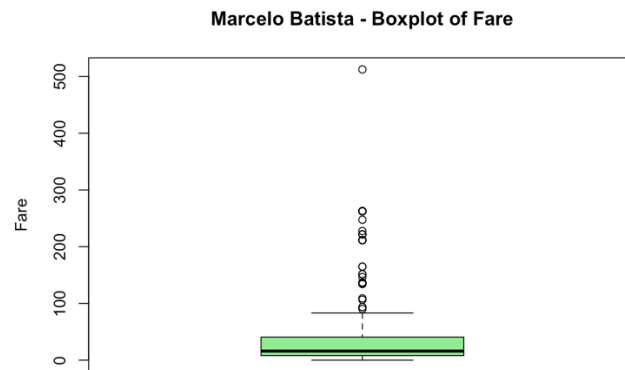
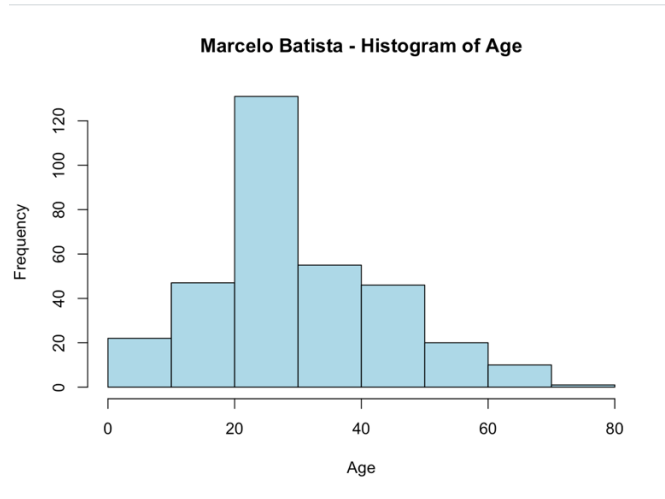
Feel free to use the table below to organize your answers. Remember to provide screenshot of your R codes. (Hint: lab 2 and assignment 1).

Names of the variable	Description
PassengerID:	ID number of each passenger
Survived:	0 = no, 1 = yes
Pclass	Ticket class (1, 2, 3)
Sex:	Male or Female
Age:	Age of passenger
Sibsp:	Number of siblings/spouses
Parch:	Number of parents/children
Ticket:	Ticket number
Fare:	Ticket price
Cabin:	Cabin number
Embarked:	Where passenger got on the ship

These variables describe passenger information such as demographics, ticket details, and family relationships. Some variables are numeric, while others are categorical.

3. Visualization

Generate at least two visualizations on the variables you choose. You can do frequency plot, relationship plot, histogram, boxplot



Describe your two plots:

3.1. What trends do you see?

In the histogram of Age, most passengers seem to be young adults, with many ages concentrated around the 20s and 30s. There are fewer very young children and fewer elderly passengers.

In the boxplot of Fare, most fares are relatively low, but there are a few very high fares. This suggests that the fare distribution is right-skewed.

3.2. Any outliers you notice?

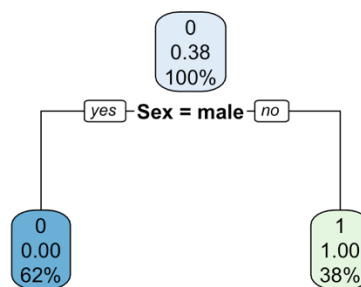
Yes. The Fare variable clearly has outliers because some passengers paid much more than the rest. The maximum fare of 512.3292 is much higher than the median fare of 16, so those high-fare passengers appear to be outliers.

3.3. Please provide screenshots of the plots with your name as the title (refer to lab 2).

4. Model1: Decision tree model

Please run the decision tree model and then report the result.

Marcelo Batista - Decision Tree



4.1. What is the most important predictor for surviving?

The most important predictor is Sex.

4.2. Why does this result make sense in the context of the Titanic?

This result makes sense since women are more likely to be given priority during evacuation under the “women and children first” rule. In this dataset, Sex is especially important since the data shows a perfect pattern: all females survived and all males did not survive.

Because of that, the tree will split on Sex first and it may even solve the classification immediately.

4.3. Remember to provide screenshots of both your R code and the tree model. (Hint: lab 4)

The decision tree is very simple because Sex alone predicts survival perfectly in this file. Female passengers are classified as survived, and male passengers are classified as not survived.

5. Model 2: Regression model

Regression models like numbers, we need to do some data transformations now.

5.1. Which variable/variables you want to change into numbers? How do you do that?

I would change Sex and Embarked into numeric variables. For example:

Sex: female = 1, male = 0

Embarked: create dummy variables such as Embarked_Q and Embarked_S, while C can be the reference category

Convert Sex to numeric

```
titanic$Sex_num <- ifelse(titanic$Sex == "female", 1, 0)
```

Convert Embarked to factor first

```
titanic$Embarked <- as.factor(titanic$Embarked)
```

Replace missing Fare with median

```
titanic$Fare[is.na(titanic$Fare)] <- median(titanic$Fare, na.rm = TRUE)
```

Call:

```
lm(formula = Survived ~ Pclass + Sex_num + Age + SibSp + Parch +  
    Fare + Embarked, data = titanic)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.094e-16	-6.410e-18	2.900e-19	4.230e-18	8.497e-16

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.000e+00	1.548e-17	0.000e+00	1.00000
Pclass	0.000e+00	4.579e-18	0.000e+00	1.00000
Sex_num	1.000e+00	5.878e-18	1.701e+17	< 2e-16 ***
Age	4.298e-19	2.266e-19	1.896e+00	0.05880 .
SibSp	1.861e-20	3.403e-18	5.000e-03	0.99564
Parch	1.694e-18	3.802e-18	4.460e-01	0.65620
Fare	-4.058e-20	6.010e-20	-6.750e-01	0.49998
EmbarkedQ	3.722e-17	1.298e-17	2.868e+00	0.00441 **
EmbarkedS	8.193e-19	7.071e-18	1.160e-01	0.90782

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.001e-17 on 323 degrees of freedom

Multiple R-squared: 1, Adjusted R-squared: 1

F-statistic: 3.92e+33 on 8 and 323 DF, p-value: < 2.2e-16

Warning message:

In summary.lm(reg_model) :

essentially perfect fit: summary may be unreliable

5.3. Run the regression analysis and write out the regression equation. Feel free to run multiple models in your analysis. But you need to choose one to present here.

Survived = 0 + 1(Sex_num)

5.4. Is your regression model good? Why or why not?

The model shows a perfect fit with an R-squared of 1. However, this isn't realistic, and the warning says the results may not be reliable. This happens because the dataset has a perfect pattern where all females survived and all males did not.

5.5 Which predictor variable is the most significant in your model? Why?

The most important variable is **Sex_num** since it has a very strong effect and a very small p-value. It fully explains survival in this dataset.

5.6. Remember to provide screenshots of both your R code and the regression model output (Hint: lab 5 and 6).

Data Dictionary:

survival - Survival (0 = No; 1 = Yes)

Pclass - Passenger Class (1 = 1st; 2 = 2nd; 3 = 3rd)

sex - Sex

age - Age

sibsp - Number of Siblings/Spouses Aboard

parch - Number of Parents/Children Aboard

ticket - Ticket Number

fare - Passenger Fare

cabin - Cabin

embarked - Port of Embarkation (C = Cherbourg; Q = Queenstown; S = Southampton)

Part II: AI-Assisted Analysis and Reflection

1. Use AI to complete the same tasks in Part I.

I used AI to help analyze the Titanic dataset by asking for help with descriptive statistics, visualizations, and models like decision trees and regression. AI helped generate explanations, suggest R code, and interpret the results.

2. Provide The exact prompts you used

Here are the prompts I used:

- “Analyze this Titanic dataset and describe the main variables and any missing values or patterns.”
- “Give me descriptive statistics and explain what they mean in simple terms.”
- “Create two visualizations for this dataset and explain trends and outliers.”
- “Run a decision tree model and explain the most important variable.”
- “Run a regression model and explain which variable is most significant.”

3. Now evaluate the AI output

3.1. Are there any challenges you encounter during the process?

One challenge was that AI sometimes gave general answers based on the original Titanic dataset instead of focusing only on this specific dataset. I had to double-check the results in R to make sure they matched my data.

3.2. Do you like the results from AI output?

Yes, I liked the AI results because they helped me understand the analysis faster and gave clear explanations. However, I still needed to verify the results myself.

4. Compare your analysis with AI analysis, did AI produce the same conclusions as you do?

The AI analysis was very similar to my own results. Both showed that Sex is the most important variable and that Fare has outliers. Both also identified that the dataset has an unusual pattern where survival is perfectly determined by sex.

5. Which approach was faster? Which helped you understand the analysis better?

Using AI was faster because it quickly generated explanations and ideas. However, doing the analysis myself in R helped me understand the data better because I had to run the code and interpret the results step by step.

6. What is one thing AI did well?

AI did a good job explaining the results in simple terms and helping me understand the models, especially the regression and decision tree.

7. What is one thing AI did not do well?

AI sometimes gave answers that were too general and not fully based on this dataset. It required me to check the results carefully to make sure they were correct.

8. Based on this experience, how would you use AI responsibly in future data analysis tasks?

In the future, I would use AI as a support tool to help generate ideas and explain concepts, but I would still perform the analysis myself. This ensures that the results are accurate and that I fully understand the process.